



The Future of AI in Microsoft Fabric: Data Agents and Beyond

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Microsoft Fabric Roadshow



Agenda

Why Data Agents?

Customer story: Porsche Cup

Data agents: What's new?

Customer story: Coca-Cola

Architectural musings 

Data Science: What's new?

Roadmap

Why Data Agents?

To ground your data insights on Enterprise Knowledge

Consumers



End-users



Agents



Applications



Developers



Fabric



Copilot Studio



AI Foundry



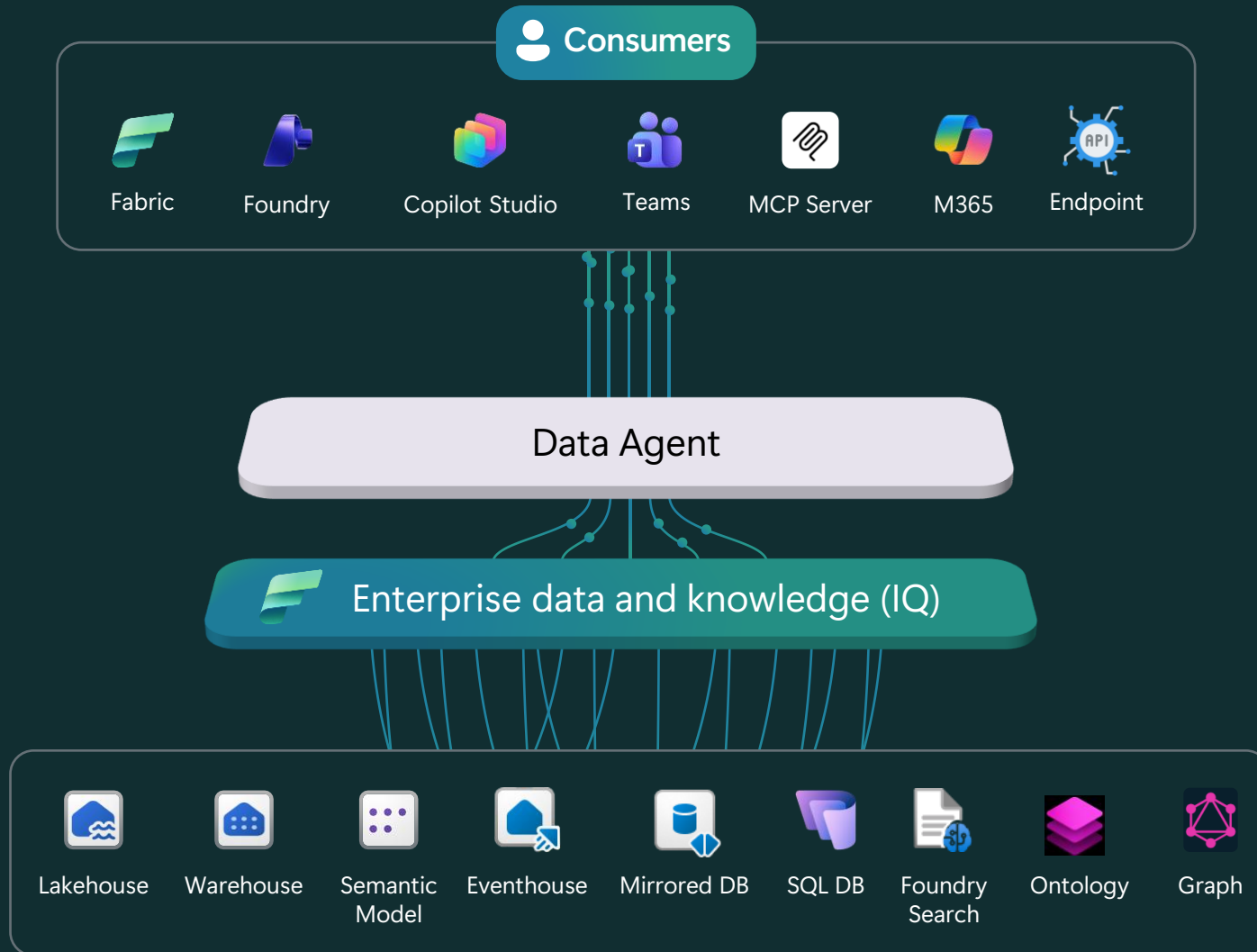
M365



API
endpoint/MCP

Data agents

Fabric data agents



A **virtual analyst** that allows users to interact with and gain insights from enterprise data and knowledge in Fabric

New

Announcing **General Availability of Fabric data agents!**

New

Data agent prompts now logged in **Purview** for **auditing and governance**

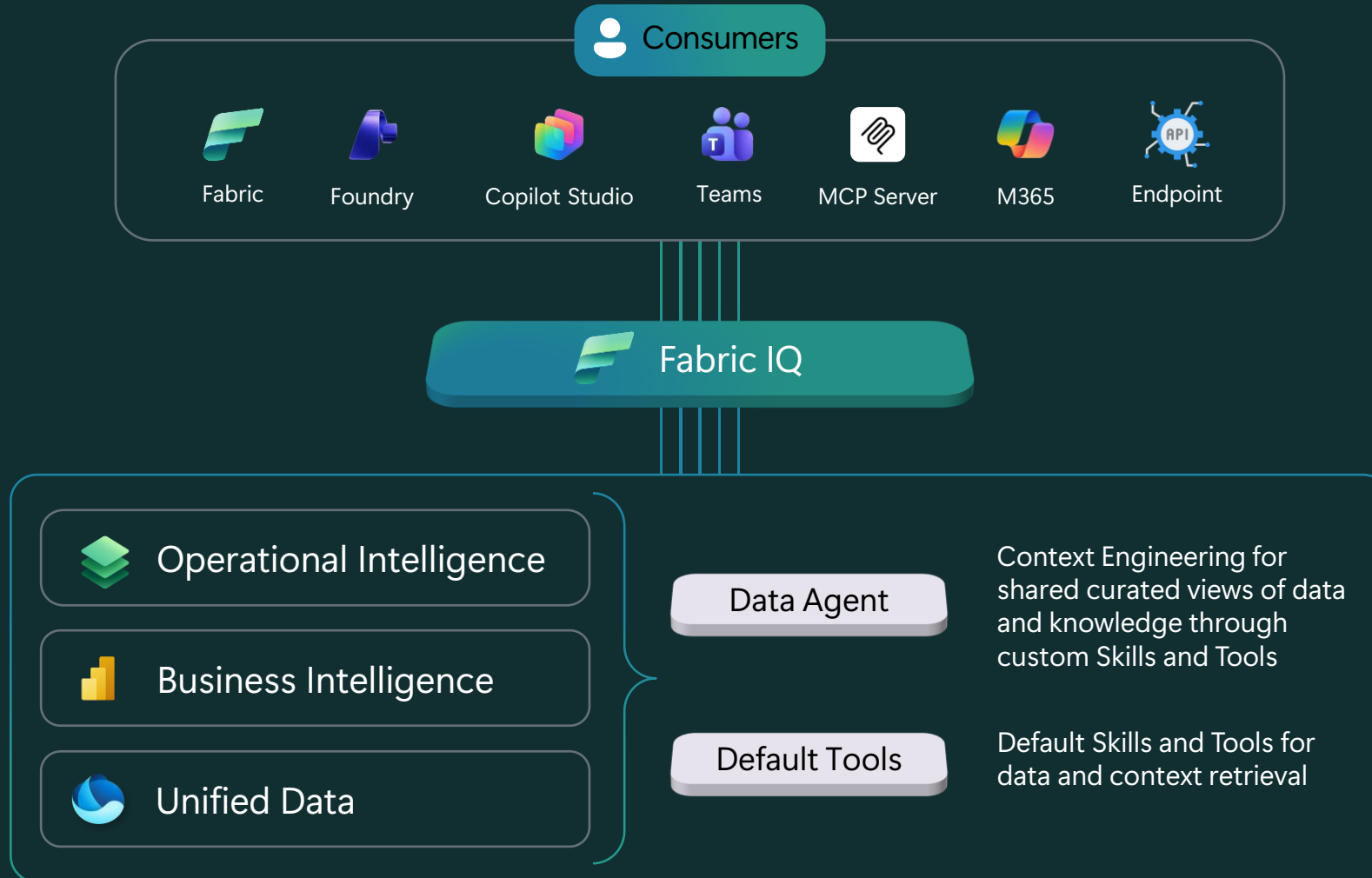
New

Graph supported as a new source for better reasoning across related entities

New

Support for **SQL Views**, and **SQL and KQL Functions**

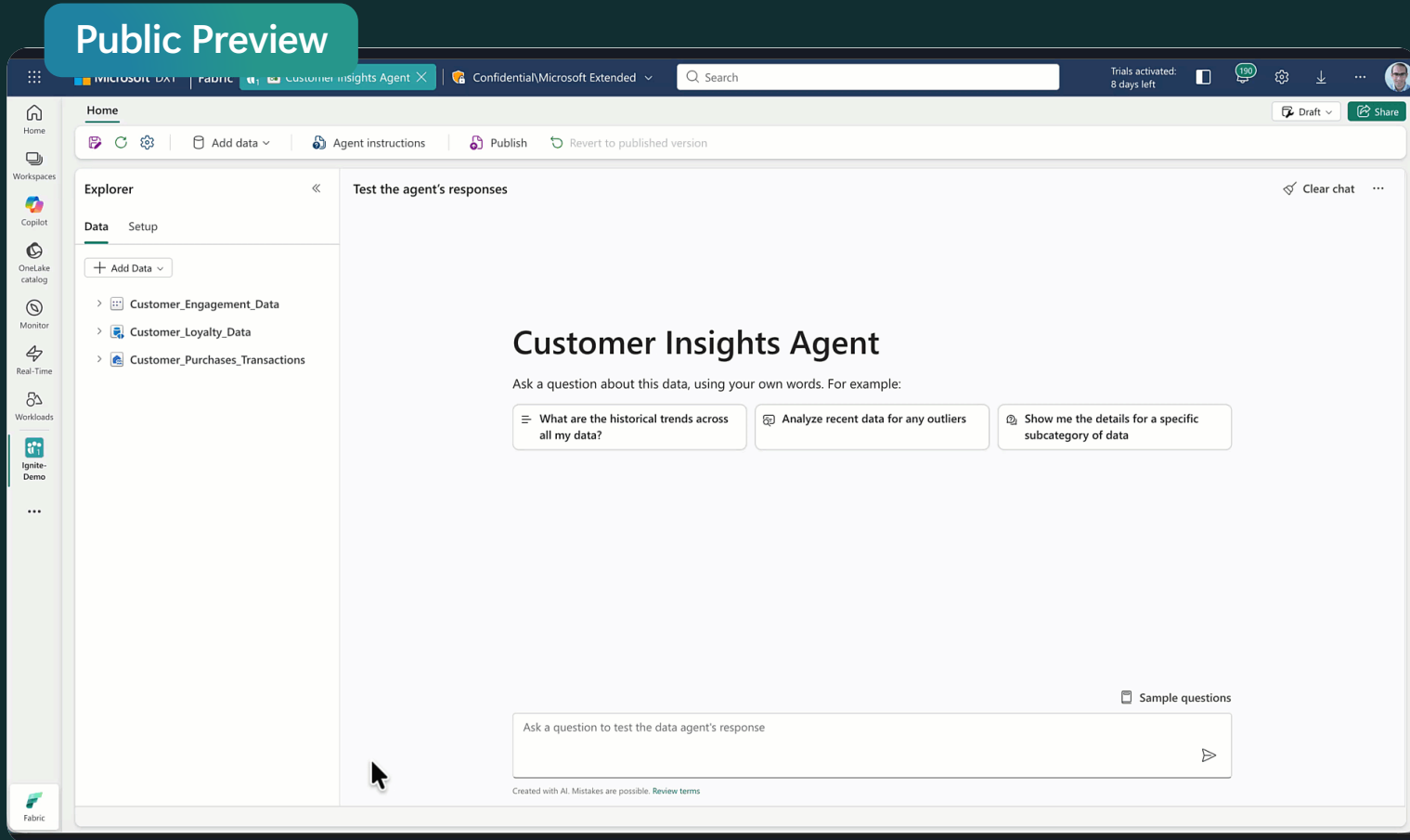
Grounding AI agents on Fabric IQ



Consuming Data Agents

Fabric data agent integration with M365

Enrich Microsoft 365 agents with Fabric data intelligence



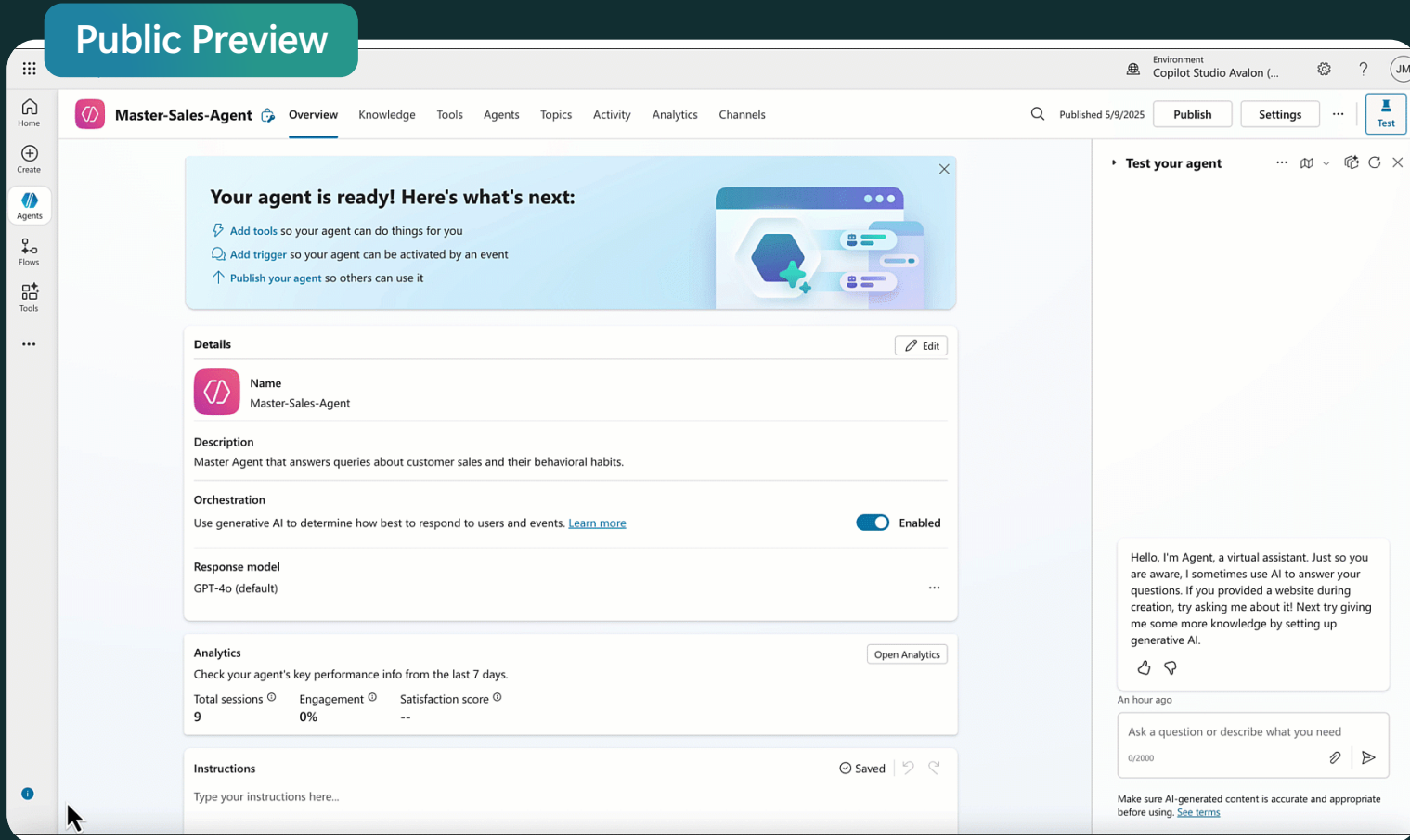
Fabric data agents can reason over and synthesize data in OneLake for insights

AI creators can enrich M365 Copilot with Fabric data agents' data domain expertise

Security permissions, including RLS and CLS, are always respected

Fabric Data Agent integration with Copilot Studio

Enrich custom copilots with Fabric data expertise

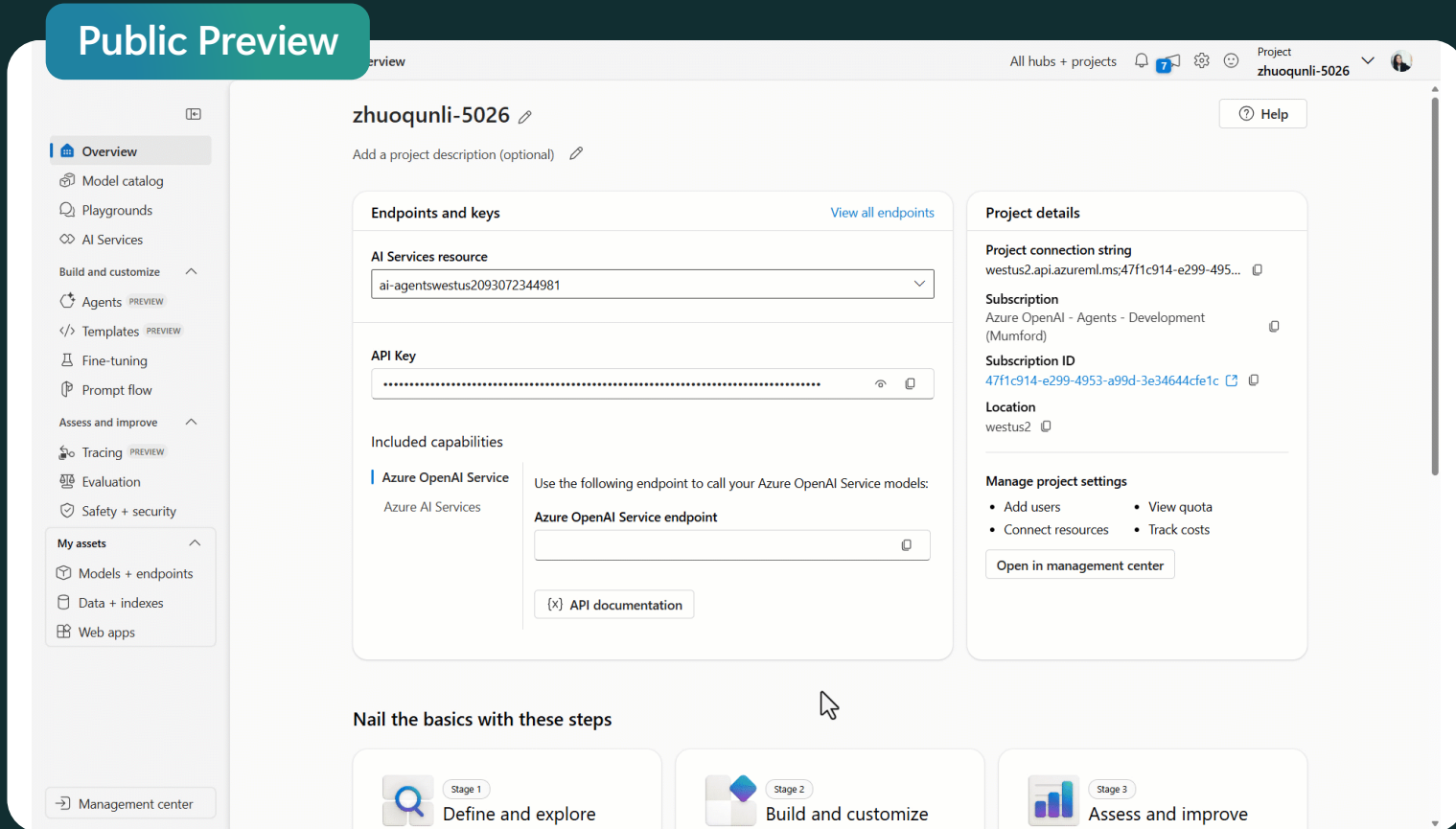


Fabric data agents can reason over and synthesize data in OneLake for insights

AI creators can enrich custom copilot agents in Copilot Studio with Fabric data agents' expertise

Security permissions, including RLS and CLS, are respected

Fabric data agent integration with Azure AI Foundry

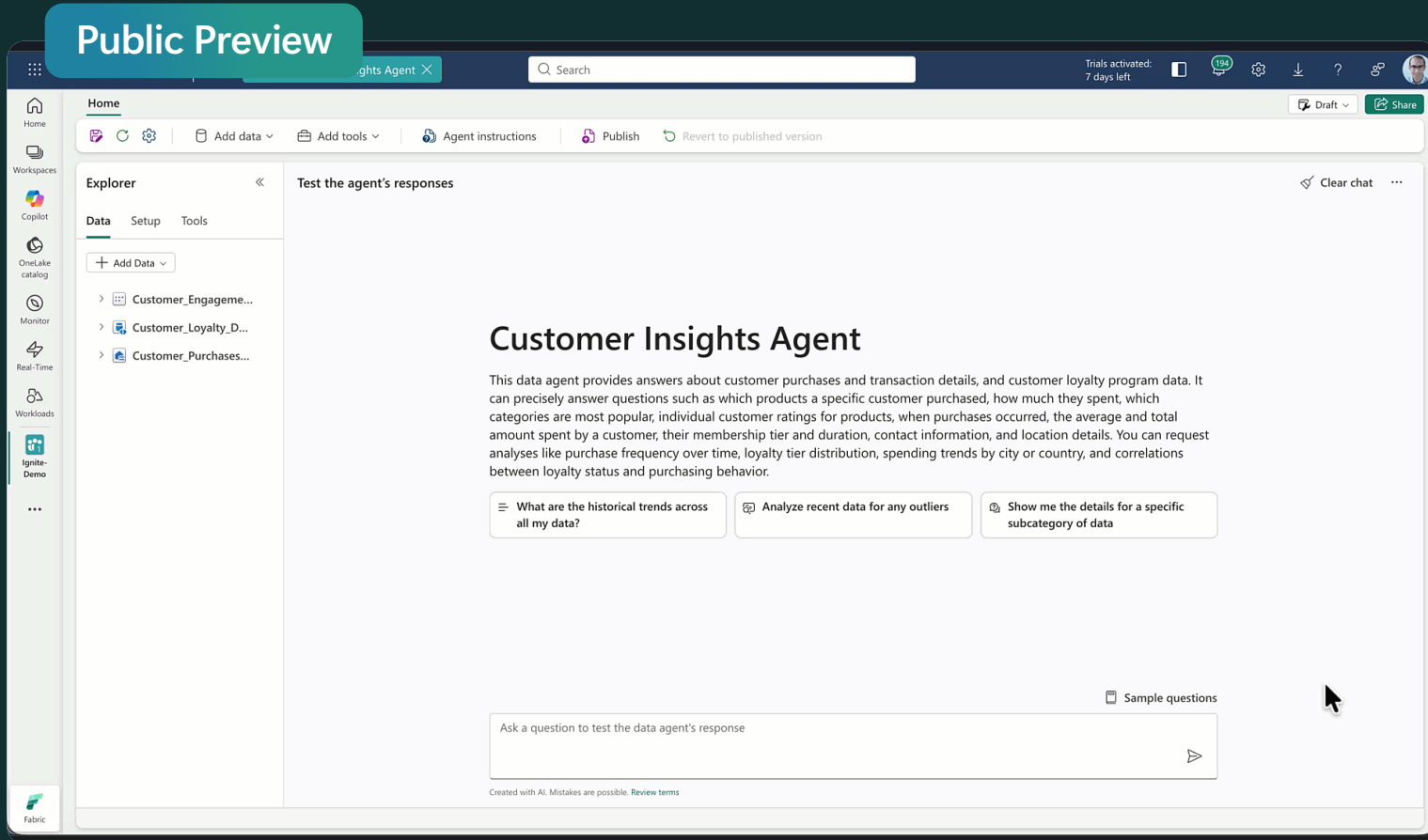


Build accurate, relevant, and context-aware multi-agent solutions

Securely ground AI agent outputs in enterprise knowledge using Fabric data agents

Security permissions, including RLS and CLS, are respected

Data Agent MCP Server



A managed MCP Server for first class access to enterprise knowledge

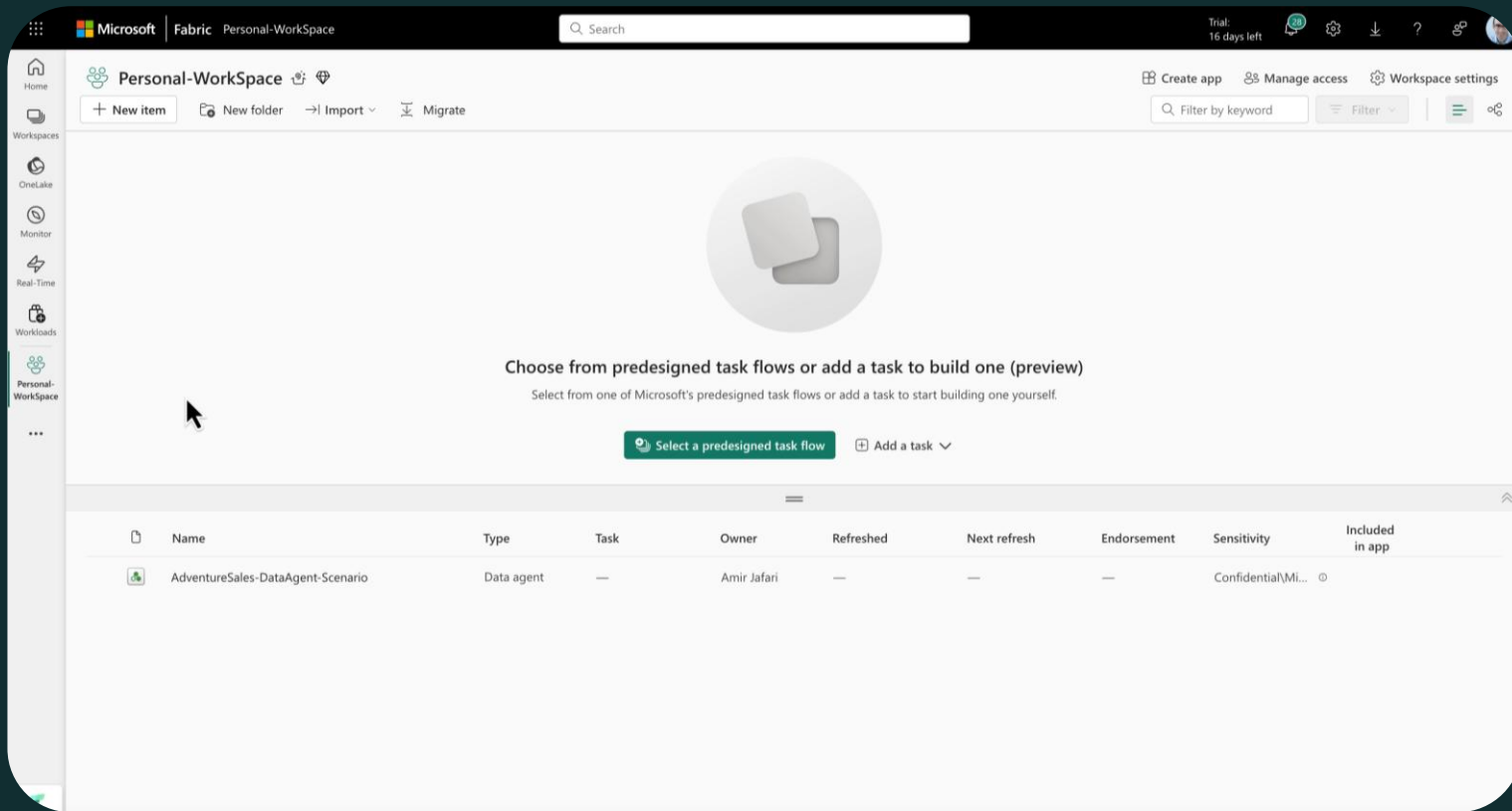
Enables users to consume Fabric data agent from other AI systems using MCP

Gain on-demand insights on top of enterprise data in Fabric, ensuring secure, and auditable data access

Data agents

What's new and upcoming?

General Availability : Data Agent in Microsoft Fabric



Create a powerful data expert tailored to your data domain by including data sources such as **Data Warehouse, Lakehouse, Semantic models, Eventhouse, SQL databases, Mirrored databases**

Configure your data expert using **agent instructions, data source instructions, and example queries**

Share data agents with other creators and consumers in **Fabric**

Home

Workspaces

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

Zava Real-Time ...

Fabric

Microsoft DXT

Fabric

ZavaTradeExpert X

Confidential\Microsoft Extended

Search

Trials activated: 21 days left

Home

Draft

Share

Add data

Add tools

Build agent with AI

Test data agent

Agent instructions

Publish

Revert to published version

Explorer

DataSetupTools

Add Data

CustomerActivity

LoanFinanceOntology_graph...

RetailOntology

Test the agent's responses

Clear chat

ZavaTradeExpert

ZavaTradeAgent is an AI assistant that helps users analyze and explore retail, financial and lending data related to supply chain and financial loans, borrowers, payments, and risk events. It answers questions about product demand as well as loan performance, borrower behavior, underwriting decisions, and delinquency patterns by connecting information across financial entities to provide clear, data-driven insights.

What are the historical trends across all my data?

Analyze recent data for any outliers

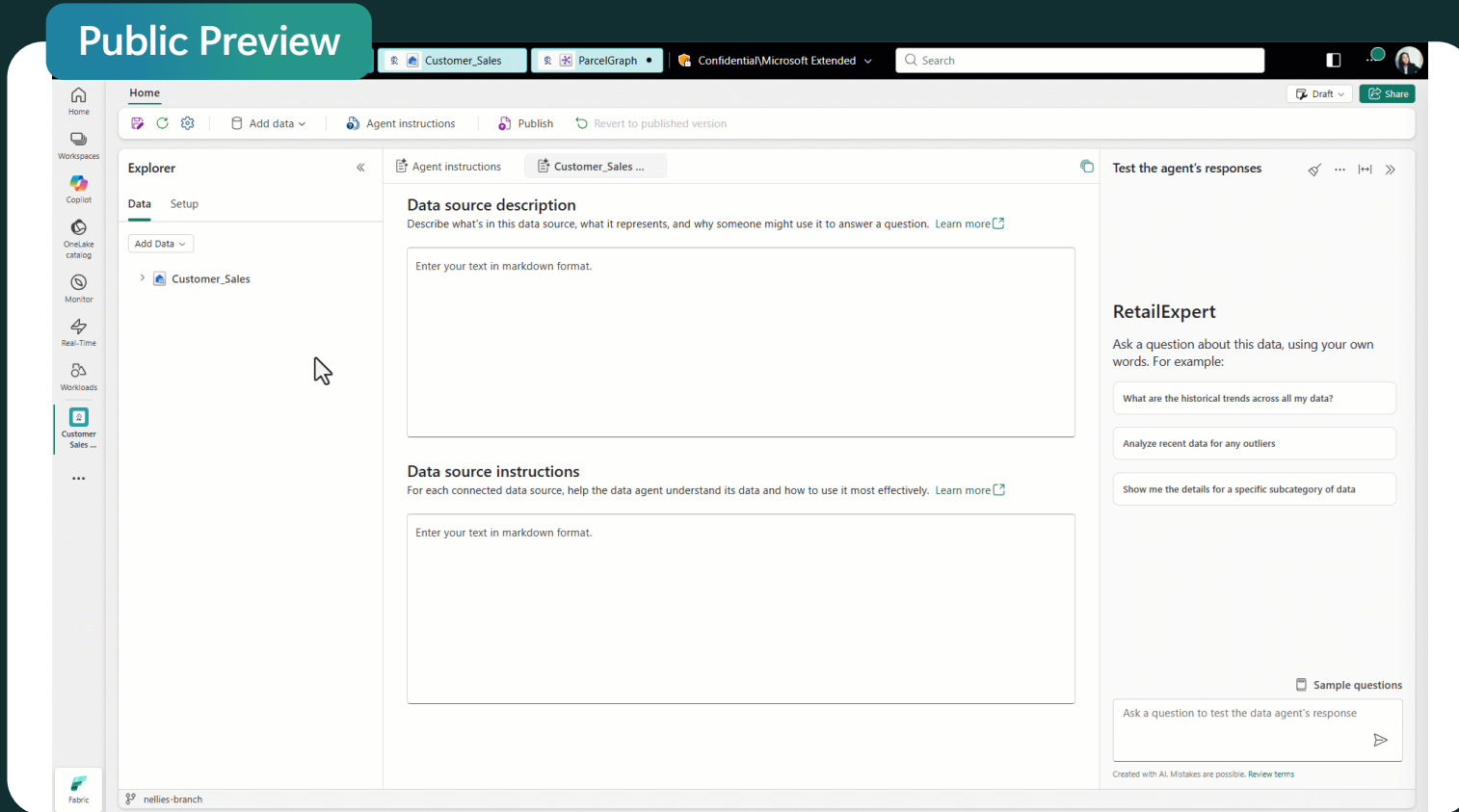
Show me the details for a specific subcategory of data

Sample questions

Ask a question to test the data agent's response

Created with AI. Mistakes are possible. Review terms

New Data Source: Graph DB

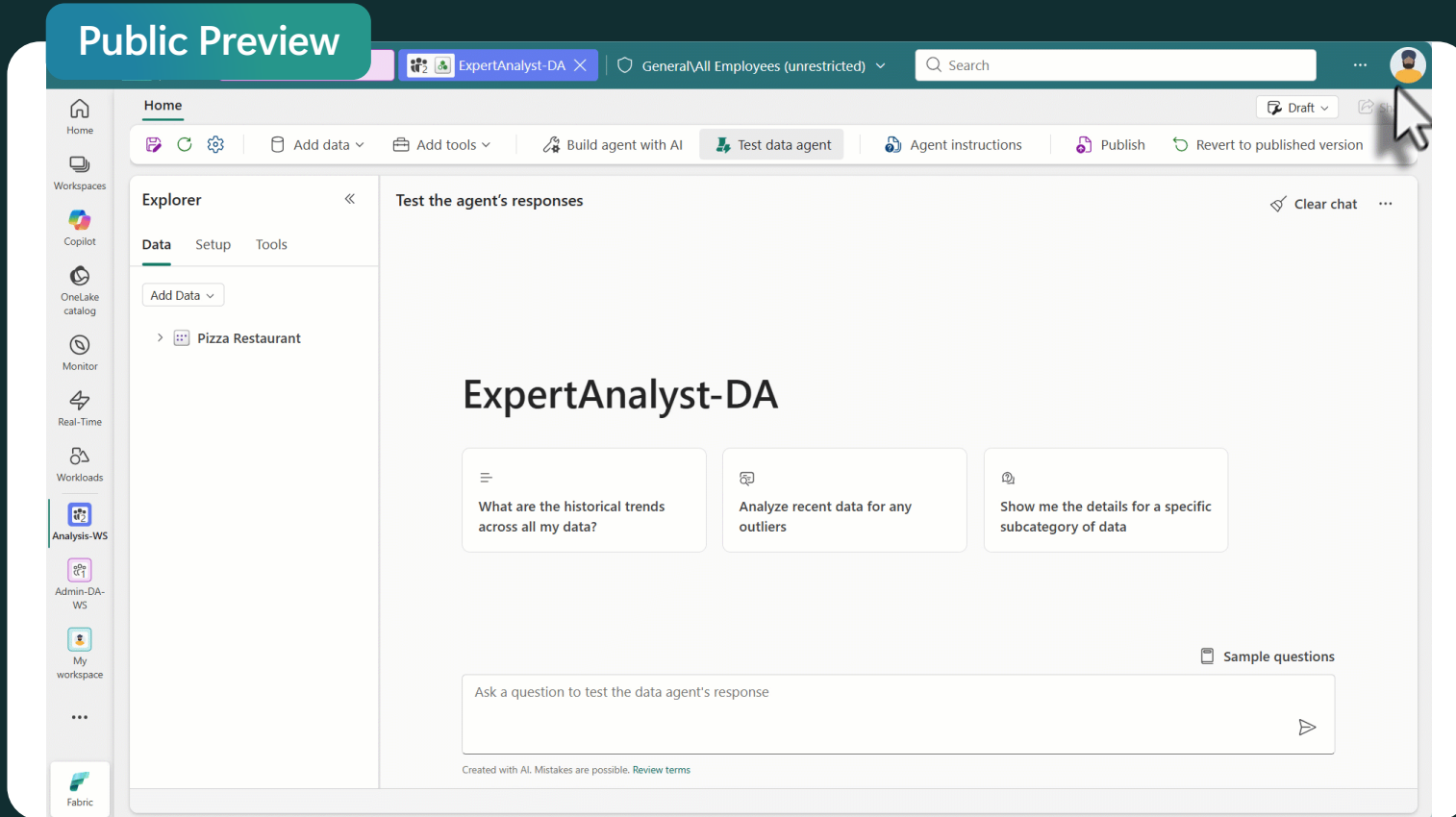


Graph Models as new source in Fabric Data Agents. NL-to-GQL tool translates user intent into executable graph queries

Relationship-Aware Reasoning Every answer is inspectable by GQL query tracing explicit paths across nodes, edges, and relationships

Enhance Graph's capabilities with Business context: Leverage Data Agent's configurations to further enhance and contextualize NL-to-GQL results

Purview Integration: Data agent Audit Logs

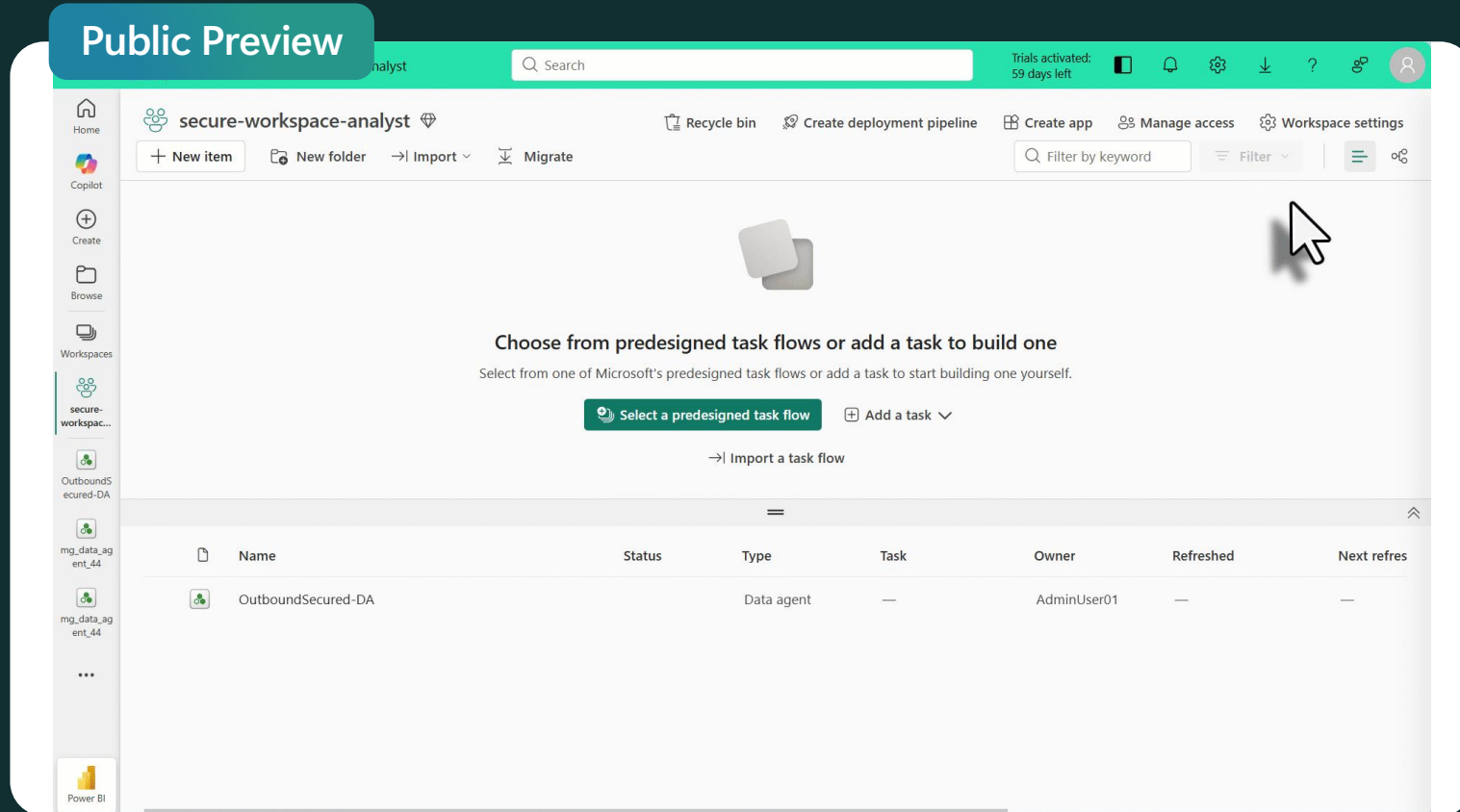


As a Fabric Admin, see what users **are asking** Data Agent

Monitor interactions and **sensitivity label references** for each AI agent

Captures **user id, capacity, accessed resources, prompt, and agent response including query**

Outbound Access Protection support with Data Agent

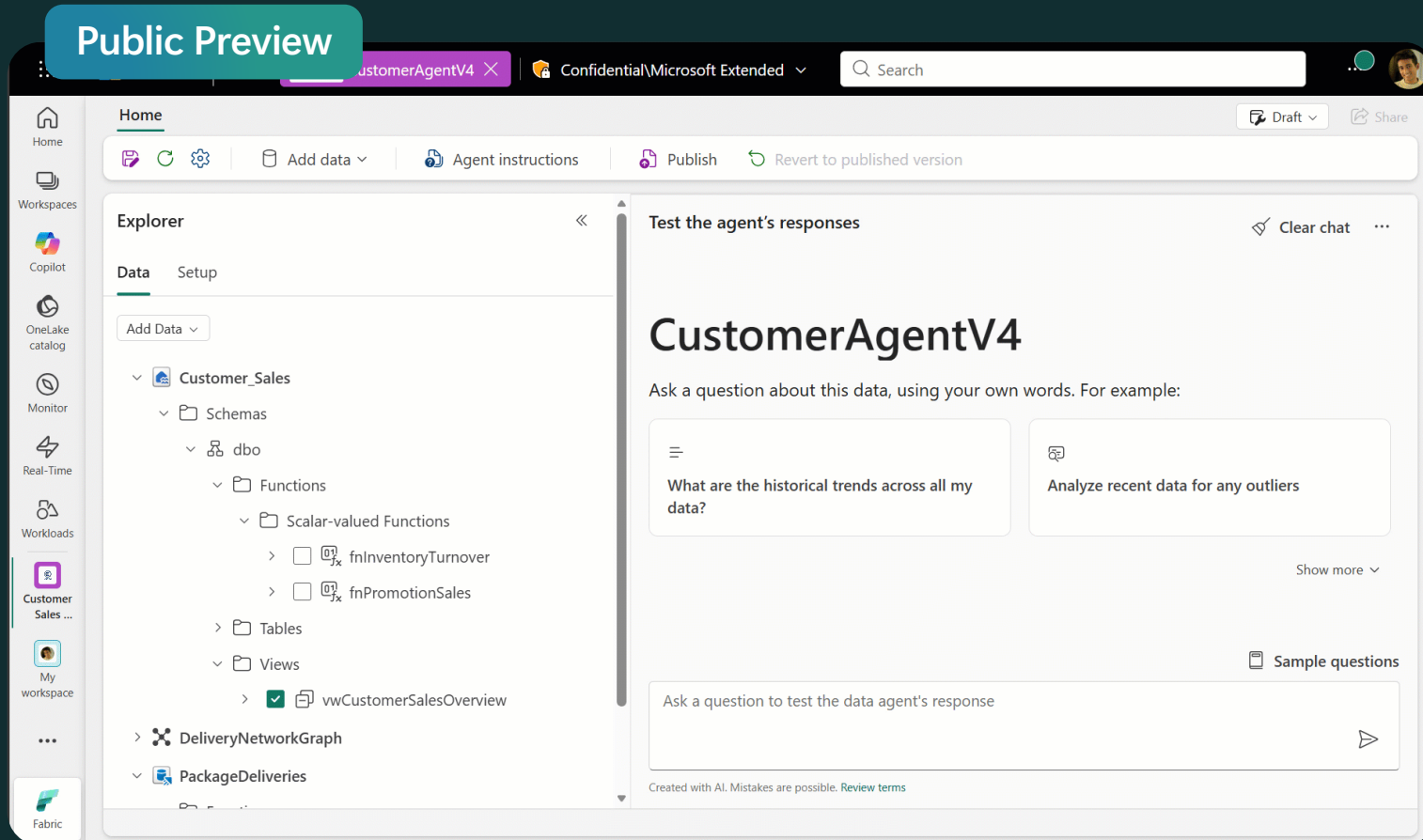


Data Exfiltration protection by **blocking all outbound traffic** from your Data Agent by default

Applied at the **individual workspace level** to enable granular control

Create **exceptions to allow specific connections** to access data sources outside the workspace

Support for SQL Views and Functions



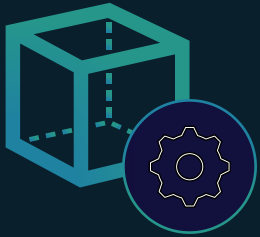
Reuse trusted **business logic, calculations, and transformations** that already exist in your databases

Ensure **consistent results**, aligned with enterprise definitions, security rules, and governance standards that are already enforced in SQL

Improve **NL2SQL accuracy** while handling complex joins, filters, and calculations transparently

Service Principal Authentication

Coming soon



App/Service
Principal



Access
Token



Data
Agents

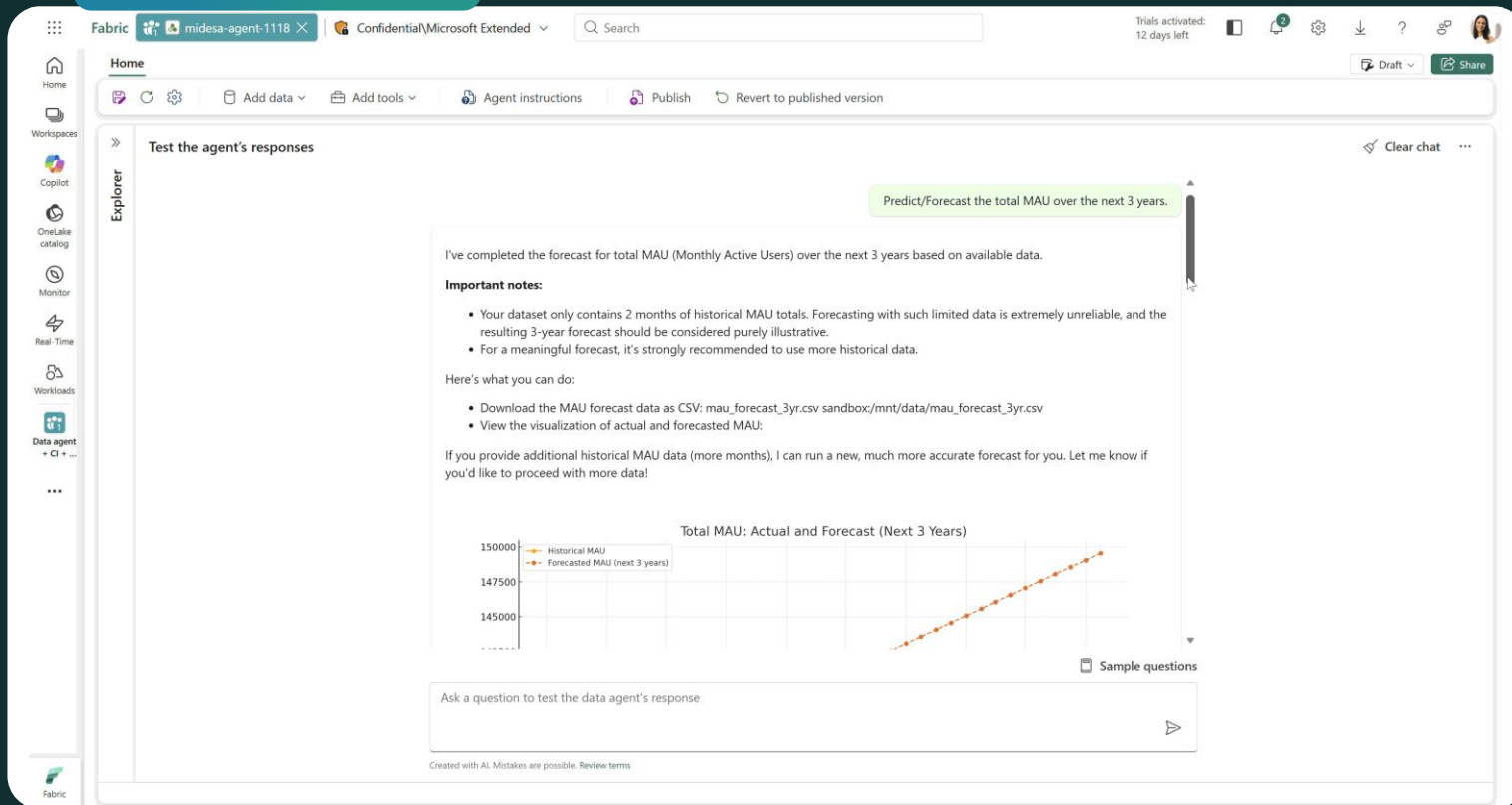
Enable secure, app-based access to Fabric data agents without requiring user sign-ins

Connect Foundry agents powered by Fabric data agents directly to your custom applications

Use SPNs for a secure and seamless connection from your app to the Data Agent MCP server

More Sophisticated Analysis with Code Interpreter

Coming soon



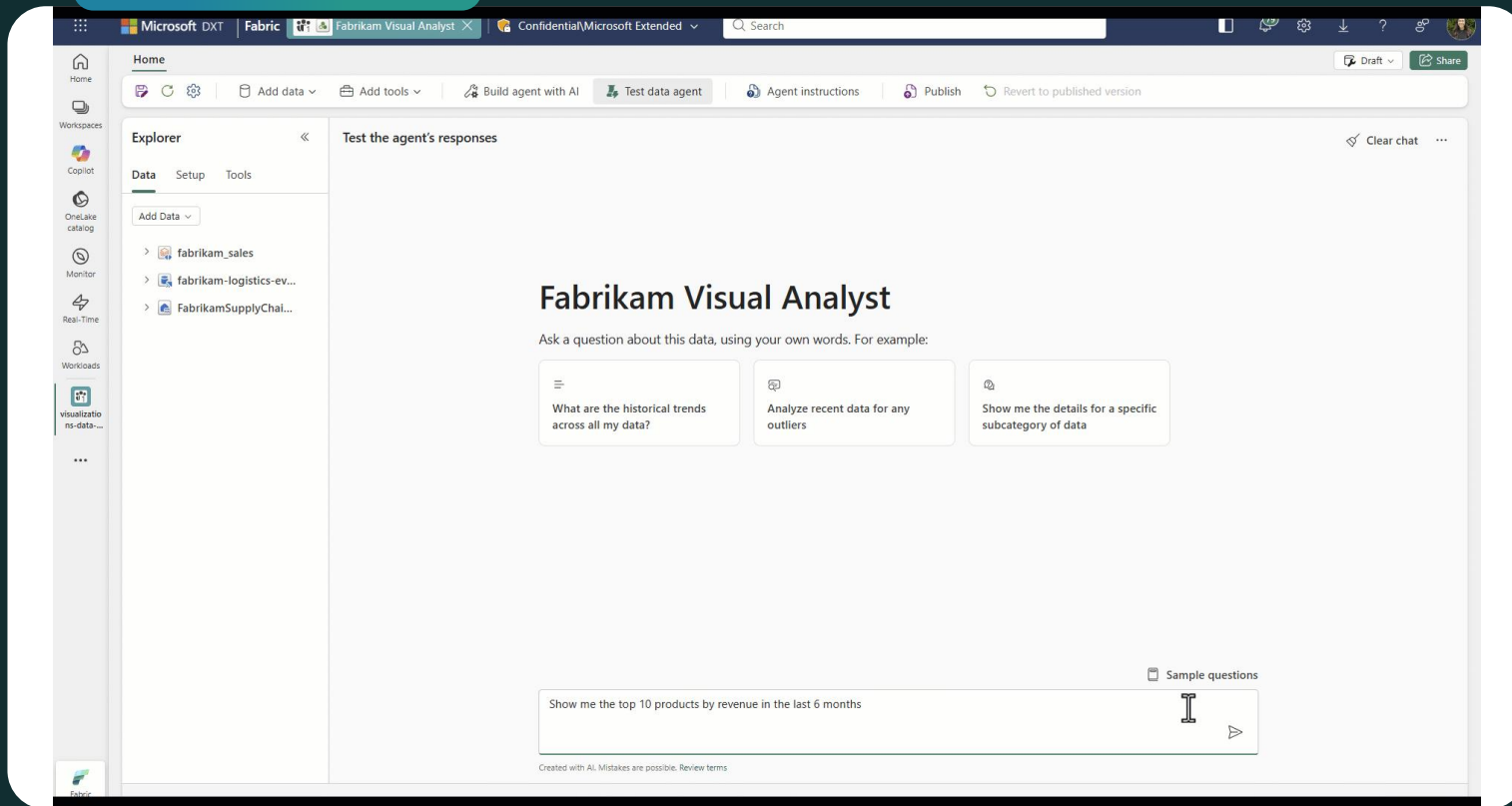
Data Agent **generates and executes Python** in real time to perform deeper, more flexible analysis

All code runs in a **sandboxed environment**, isolated from your system

Tackle **complex logic, statistics, and visualizations**, while seeing the generated code and outputs for trust and reuse

Visuals in Data Agents

Coming soon



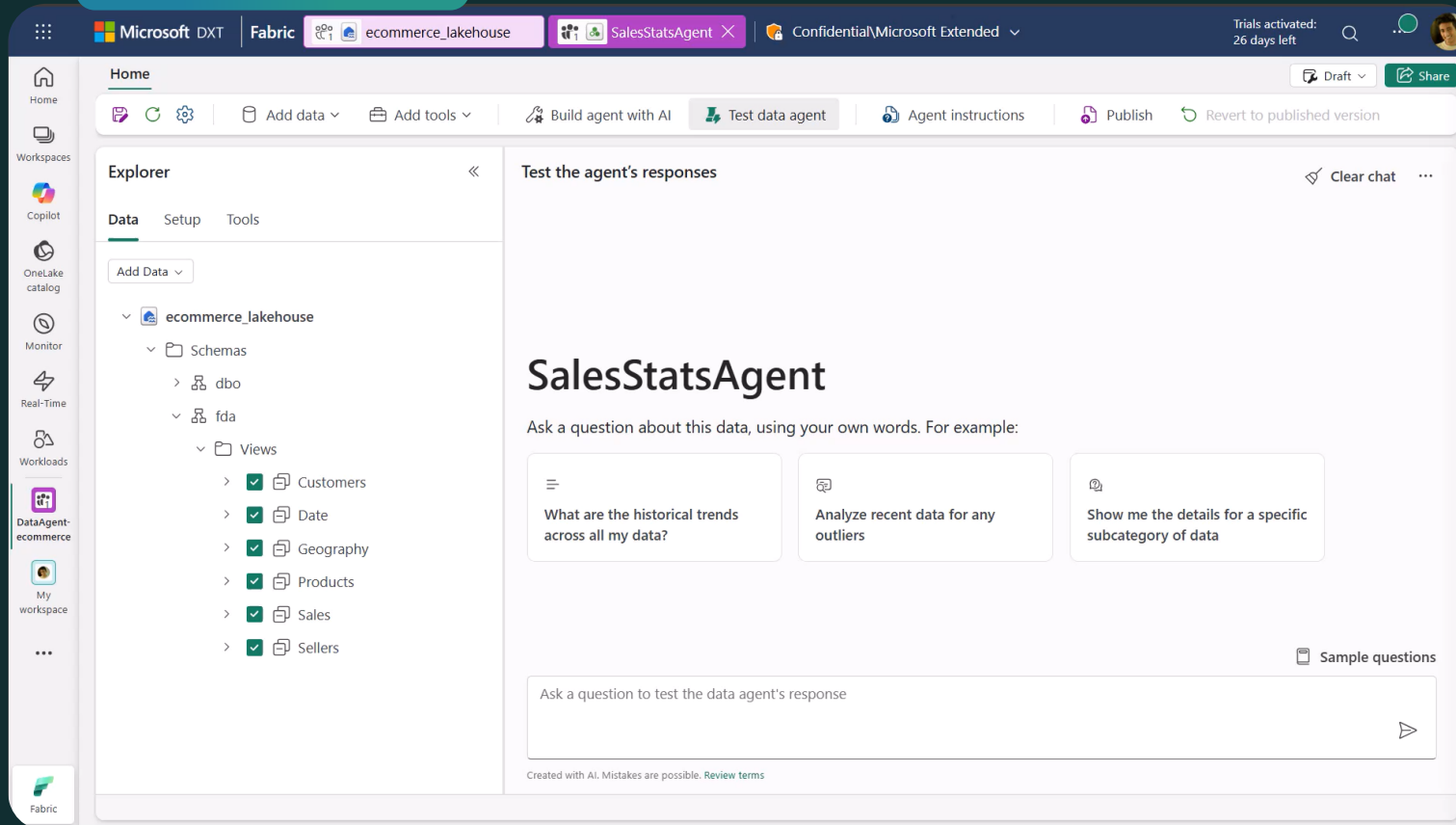
Make complex data instantly understandable with **built-in visualizations** in the Data Agent

Supported visuals include **line, bar, stacked bar, pie, scatter, area, and stacked area charts**

Charts are **interactive** — hover for **tooltips** and **select/ deselect series** to focus your analysis

Assisted Setup Mode

Coming soon



Use a collaborative approach to generate contextual configurations for data agent

Improve existing data agents using an iterative approach to refine instructions and example queries

Validates configurations using creator feedback to resolve ambiguity and conflicts

SQL Query Generation Improvements

Coming soon

The screenshot displays the Microsoft Fabric Customer Analysis Agent interface. The top navigation bar includes 'Fabric', 'Customer360', and 'Customer Analysis Agent'. The main workspace is titled 'Test the agent's responses'. A chat window shows a user prompt: 'Show top 20 VIP customers by card-not-present spend this year.' The agent's response includes a text explanation and a table of results.

The table shows the top 20 VIP customers by card-not-present spend. The columns are: customer ID, full name, email, card-not-present transaction count, and total card-not-present spend in USD.

	customer ID	full name	email	card-not-present transaction count	total card-not-present spend in USD
1	CUST0000158	Sanjay Garcia	sanjay.garcia4482@yahoo.com	1	31.14

Below the table, a list of steps completed is shown: Analyzed Customer360 Lakehouse, Row Sampling, Value Distribution, Execute SQL, and Submit Final SQL. A 'Sample questions' section is at the bottom.

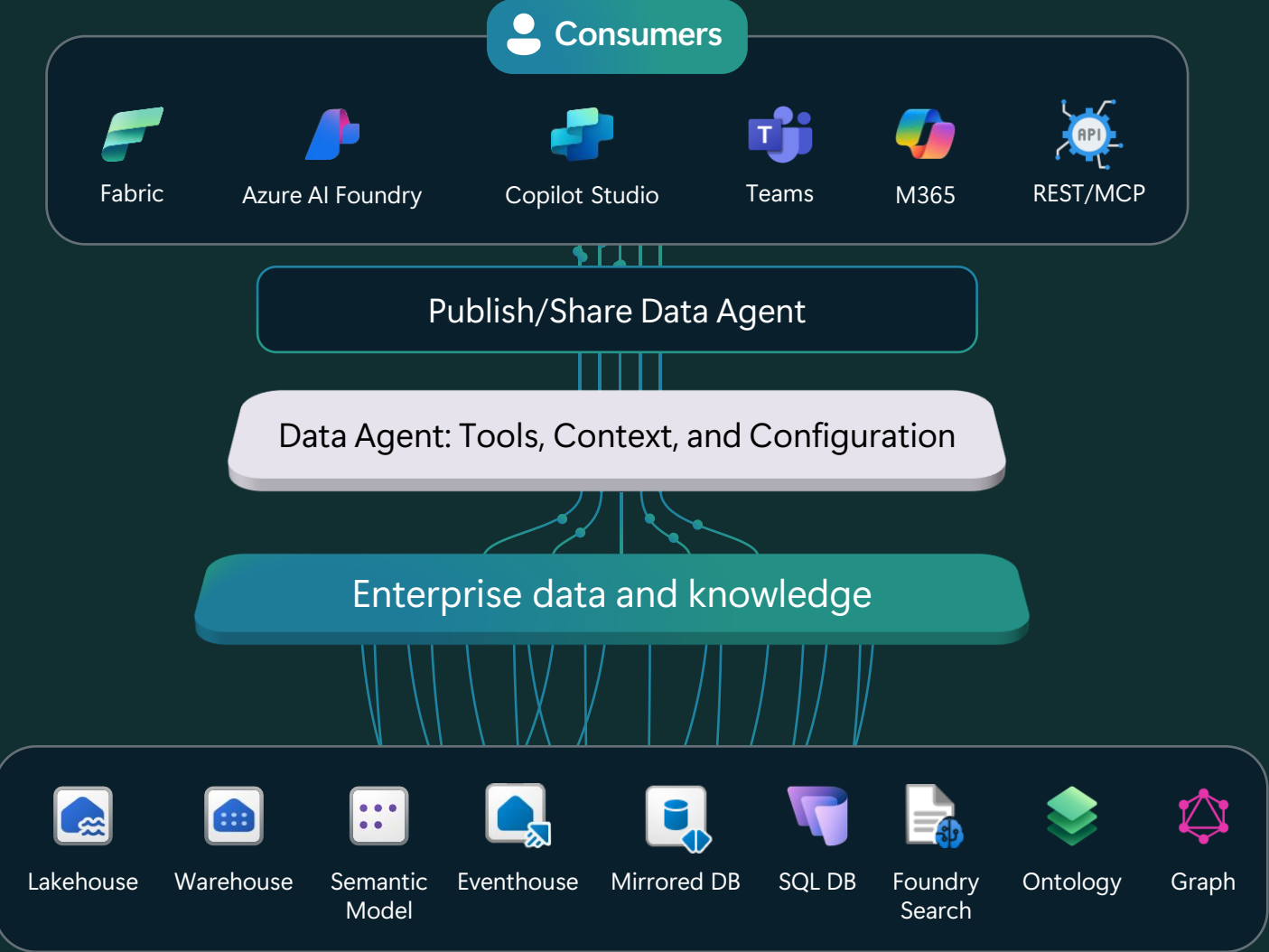
Access to **new sub-tools** to explore and generate a query

Enhanced support for **few shot following** and **filter value substitution**

New data exploration tools to assess joins and filter values for 'novel' questions

Architectural Musings

Fabric Data Agents: Technical Design



Open and Compatible

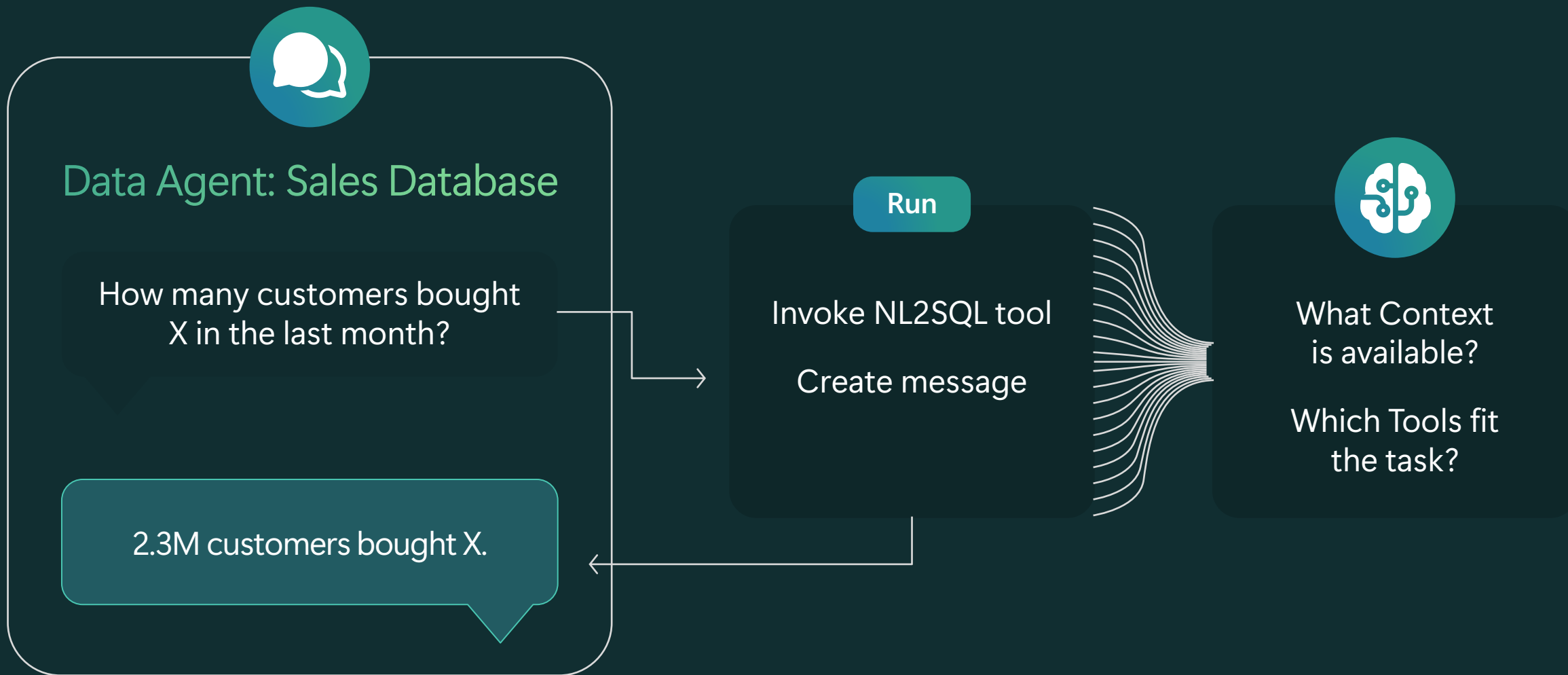
Support any Data in Fabric
Built on industry standard stacks



"Don't be weird"

Follow industry standards
Shape them where needed

Data Agents: Flow



Data Agent Components

Access APIs (Chat, MCP)

(System) Prompt

- Our not-so-secret sauce
- Your AI instructions

Tools

- One Tool per Data Source
- Code Interpreter
- Visuals
- ...

Context

- Session history
- User information
- Database metadata

MCP Server and Client In Data Agent

MCP Servers

- **Tools:** Functions for an agent to invoke
- **Resources:** Context for an agent to consider
- **Prompt templates:** For an agent to fill out and use

Data Agent

Is an MCP Server

- Any Data Agent can be invoked as a Tool

MCP Client

- Part of an Agent
- Connects the Agent to a MCP Server
- Invokes Tools, fetches Context and Templates

Data Agent

Is an MCP Client

- Every data source is a Tool
- **Future?:** BYO MCP Server with Tools

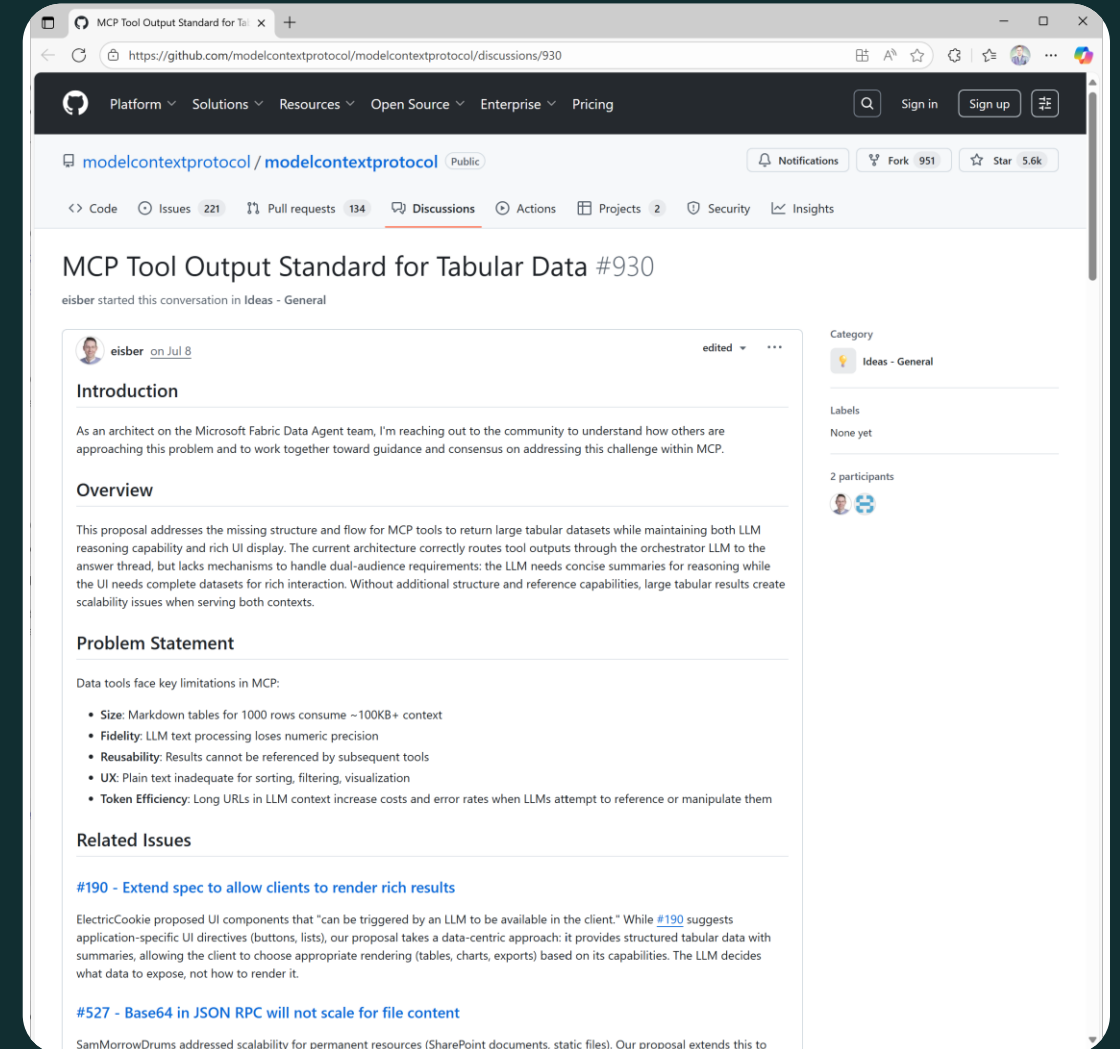
MCP for Structured Data

MCP supports flexible data representations

There is no prescription for tables

Microsoft is publishing its approach to tabular data in MCP

github.com/modelcontextprotocol/discussions/930



Yes, but what if the agent runs this CLI?

```
$> sqlcmd -Q "  
    EXEC sp_MSforeachtable 'DROP TABLE ?'  
    "
```

MCP Servers are APIs for Agents



MCP Servers expose APIs appropriate for Agents

Simplified and focused to the use case

Reduced security exposure

Example: Data Agent's SQL Tool does not allow changes



MCP Servers distinguish human from Agent actions

Important for audit logs
(see also: Purview integration)



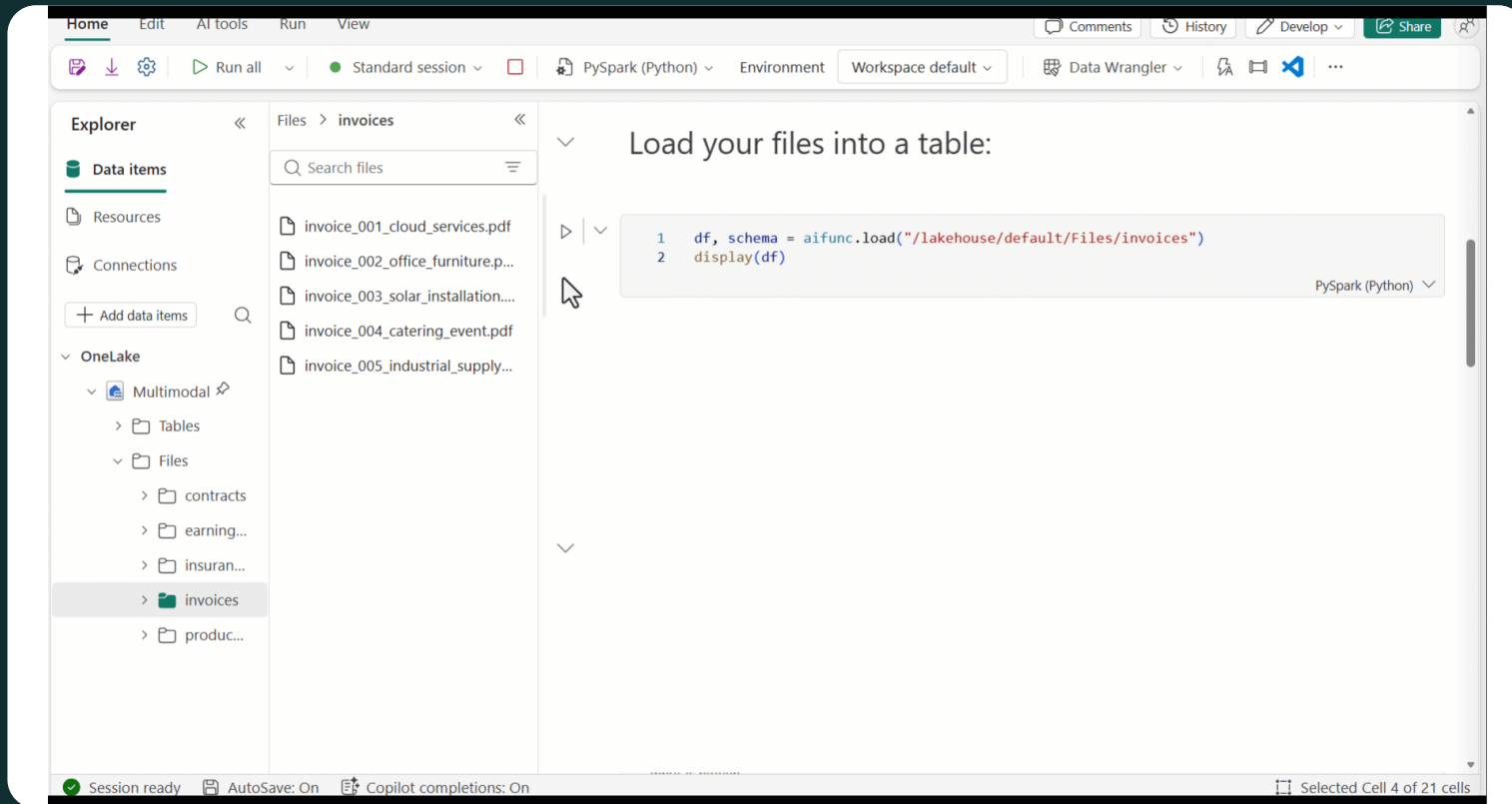
However, code and CLI access give agents infinite abilities

We "just" have to make it safe

Stay tuned 😊

Data Science

AI Functions: Multimodal Support

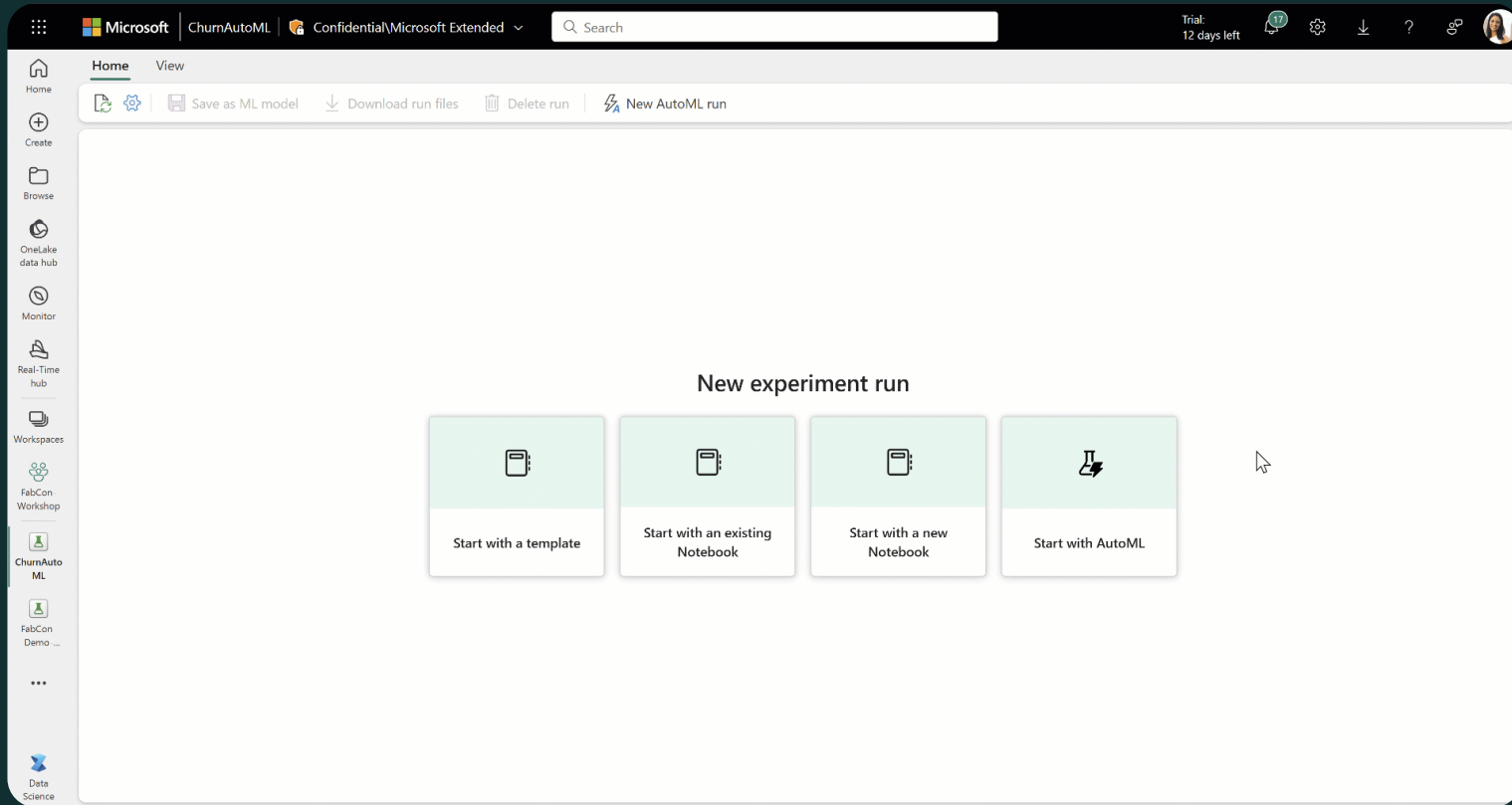


AI functions now accept **images**, **PDFs**, and **text files** as input in addition to plain text

Turn a folder of **unstructured files** into a **structured table** in a single call with **aifunc.load**

Uses the same **familiar AI functions**: **ai.classify**, **ai.summarize**, **ai.extract**, **ai.translate**, and more

Data Science: AutoML GA

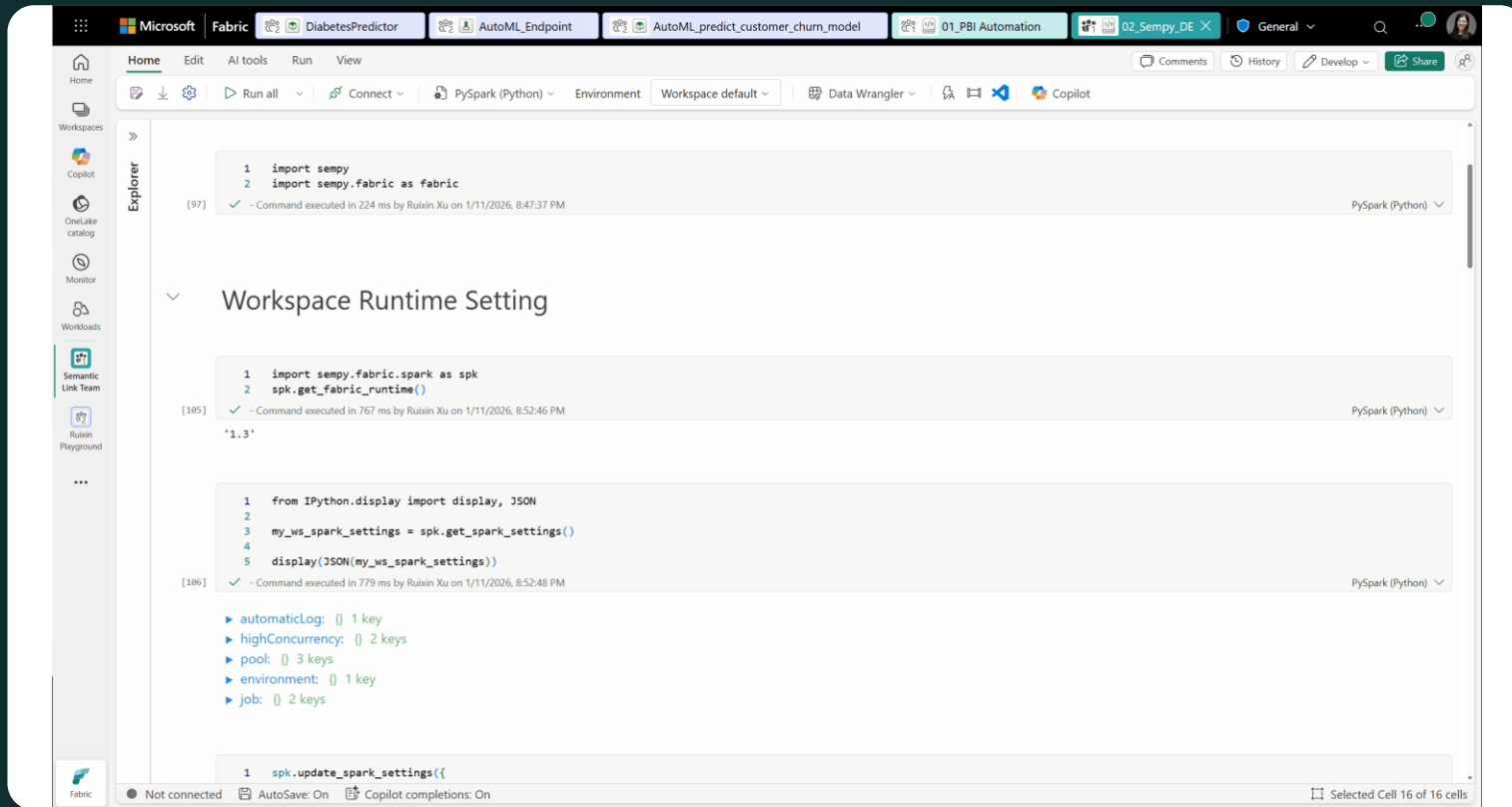


Built-in AutoML library to automatically select models and tune hyperparameters, reducing manual experimentation

Spark-powered parallel trials to scale AutoML runs across distributed compute for faster results

Low-code training experience that enables users to configure, run, and deploy models in just a few clicks

Semantic Link: expand to end-to-end workspace operations



Expanded Semantic Link management, with new modules covering Lakehouse, Reports, SQL Endpoints, and Spark

End-to-end workspace operations via code, enabling teams to configure SQL & Spark programmatically, manage data assets, reports, models,

More reliable and seamless automation, with service principal supported, improved item resolution and key bug fixes

Data Science Roadmap

Roadmap

Coming soon

Data Science

- MLFlow 3 :
 - Logged models as first class objects
 - Built-in tracing for ML observability
- Log models across workspaces and from outside
- Prebuilt content understanding
- Feature Store
- DNN training with GPU accelerated nodes
- Bring your own ML runtime
- ML Ops : model alias, model registry, automation

Data Agent Roadmap

Roadmap

Private Previews

- Visualizations & Creator Assisted Workflows
- Improved Natural Language to SQL

Coming soon (May 2026 onwards)

- Improved analysis tools: Code Interpreter
- Agent Observability in Foundry
- Service Principal Authentication
- Schema Descriptions
- Creator/Consumer feedback loop
- Unstructured data in OneLake

Learn more

Get started

<https://aka.ms/Fabric/create-data-agent>

<https://aka.ms/fabric/data-agent/sdk/docs>

<https://aka.ms/data-agent-evaluation-sample>

<https://aka.ms/data-agent-evaluation-sample>

Power BI Semantic Models + Data Agents

<https://aka.ms/data-agent/semantic-model/best-practices>

Call data agent endpoints

<https://aka.ms/fabric/data-agent-python-client-sdk>



Feature request?

ideas.fabric.microsoft.com



Thank You

Microsoft Fabric Roadshow

